

SUMMARY

My approach to ecological questions is fundamentally interdisciplinary & systems-oriented. I am interested in how organisms coalesce into communities & how complex community interactions shape the structure & function of systems. I combine field ecology, molecular & computational methods, ecological statistics, & advanced visualization to investigate how biological systems respond to environmental change across scales. My work spans marine & terrestrial systems & integrates field observations, large biological data sets, analytical workflows, & cross-disciplinary collaboration into coherent ecological narratives. I am passionate about transparent & reproducible science. I use open-source tools to create data-driven workflows that communicate science more effectively.

RECENT APPOINTMENTS

- 2025
|
2022
● Research Associate
[Smithsonian Tropical Research Institute](#)  Panama
- 2022
|
2017
● STRI/Moore Foundation Postdoctoral Fellow
[Smithsonian Tropical Research Institute](#)  Panama
- 2016
|
2012
● Postdoctoral Research Associate
[Bigelow Laboratory for Ocean Sciences](#)  Maine USA

EDUCATION

- 2012
|
2006
● PhD Microbiology
[University of Wisconsin-Madison](#)  Wisconsin USA
- 2002
|
1998
● BSc Aquatic Biology, Minor in Archaeology
[University of Texas at Austin](#)  Texas USA

MARINE FIELD WORK

- 2024
|
2017
● Western Tropical Atlantic
Extensive field work in mangrove, sea grass, and coral reef systems. Primary sites located along the Caribbean coast of Panama, with an emphasis in and around the Bocas del Toro archipelago.  Panama
- 2024
|
2017
● Tropical Eastern Pacific
Five expeditions to [Isla Coiba](#) and surrounding islands to study host-associated microbiomes in mangrove and coral reef ecosystems.  Panama
- 2014
● UNOLS Chief Scientist Training Cruise
[R/V Atlantic Explorer \(cruise AE1410\)](#)  Barbados → Bermuda
10 days at sea

Jarrold J Scott

Field biologist, systems ecologist, data scientist, interdisciplinary researcher.

CONTACT

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-  ORCID
-  Google Scholar
-  web

EXPERTISE

Field: Marine (deep sea vents, coral reefs, mangroves) & Terrestrial (temperate, tropical, forest, desert).

Lab: Sterile techniques, microscopy, culturing, & isolation.

Molecular: PCR, gel electrophoresis, DNA/RNA extraction, sequencing.

Bioinformatic: Amplicon, genomic, & metagenomic analysis; various statistical methods.

Visual: Translate complex datasets into visual narratives for data exploration, hypothesis generation, & communicating results.

Publishing: Interpret data, write papers, reproducible workflows.

Other: SCUBA Rescue Diver, 4x4 Off-Road Operations, Fire Fighter I & II certification, Knots.

TOOLS

Coding Languages: R, Python

Omics: [anvi'o](#), [DADA2](#), [mothur](#), [oligotyping](#), [MED](#)

Workflows: [Quarto](#), [RStudio](#), [CSS](#), [HTML](#), [JavaScript](#)

Visuals: [Gimp](#), [Inkscape](#), [Matplotlib](#), [Plotly](#), [Leaflet](#), [ggplot2](#), [tmap](#), [Gephi](#), [Cytoscape](#).

- 2014 ● NOAA VENTS: Submarine Ring of Fire 2014 (Ironman)
[R/V Revelle \(cruise RR1413\)](#) 📍 Mariana BackArc Basin
 28 days at sea
- 2013 ● Ecology of Microbial Mats at Loihi Seamount
[R/V Thompson \(cruise TN293\)](#) 📍 Lo'ihi Seamount (Hawaii)
 28 days at sea
- 2012 ● Pyrite Nanoparticles from Mid-Atlantic Ridge Hydrothermal Vents
[R/V Knorr \(cruise KN209-02\)](#) 📍 Mid-Atlantic Ridge (Portugal -> USA)
 29 days at sea

Research cruises from 2012 to 2014 were focused on the microbial ecology of iron-oxidizing communities at deep-sea hydrothermal systems using the [ROV Jason II](#) & [AUV Sentry](#).



TERRESTRIAL FIELD WORK

- 2010 | 2008 ● Microbial Ecology of Fungus-Growing Ants
[Smithsonian Tropical Research Institute](#) 📍 Panama
 • Four expeditions to Panama • Field & lab experiments with fungus-growing ants
 • 15-month residency at STRI
- 2004 | 2001 ● Biogeography of Fungus-Growing Ants
 University of Texas 📍 Mexico & Panama
 Multiple field expeditions to understand the biogeography of fungus-growing ants & their fungal symbionts.
- 2001 | 2000 ● Molecular Ecology of Cichlids in Northern Mexico
 University of Texas 📍 Coahuila, Mexico
 Molecular analysis of cichlid fish endemic to aquifer fed pools of the Cuatro Ciénegas Basin.
- 2000 ● Mayan Archaeological Surveys
 University of Texas 📍 Northwestern Belize
 Regional scale surveys of Mayan archaeological sites in lowland tropical rain forests.

A lot of my field experience in terrestrial systems is on fungus-growing ants in the Neotropics.



OTHER TRAINING & CERTIFICATIONS

- 2015 ● Complex Systems Summer School (CSSS)
[Santa Fe Institute](#) 📍 New Mexico USA
 4-week intensive course on complex systems.
- 2014 ● UNOLS Chief Scientist Training Cruise
 The University-National Oceanographic Laboratory System 📍 Caribbean.
 2-week course on how to effectively plan and run multi-disciplinary research & education cruises.
- 2013 ● Fire Fighter I & II. NFPA 1001-2006
 Southern Maine Community College 📍 Maine USA
 Year-long training course for Fire Fighter I & II Certification.

[2015 CSSS proceedings.](#)

[Final report from the 2014 UNOLS training cruise.](#)

- 2007 ● **Microbial Diversity Course**
Marine Biological Labs  Massachusetts USA
6-week intensive course. Cultivating, & isolating diverse microbes. Molecular & computational analyses.
- 2000 ● **Archaeological Field Techniques**
The Programme for Belize Archaeological Project  Belize
Intensive field course on Mayan art, architecture, & iconography.

Learn more on the [course website](#).

Learn more on the [course website](#).



SELECT PEER REVIEWED PUBLICATIONS

- 2023 ● **Contrasting drivers of abundant phage and prokaryotic communities revealed in diverse coastal ecosystems.**
[ISME Communications 3, 127 \(2023\)](#) 
Weinheimer AR, Aylward FO, Leray M, **Scott JJ**
- 2022 ● **Microbial diversity declines in warmed tropical soil and respiration rise exceed predictions as communities adapt.**
[Nature Microbiology 7, 1650–1660 \(2022\)](#)
Nottingham AT, **Scott JJ**, Saltonstall K, Broders K, Montero-Sanchez M, Püspök J, Bååth E, Meir P.
- 2022 ● **The gut microbiome variability of a butterflyfish increases on severely degraded Caribbean reefs**
[Communications Biology 5, 770 \(2022\)](#) 
Clever F, Sourisse JM, Preziosi RF, Eisen JA, Rodriguez Guerra EC, **Scott JJ**, Wilkins LGE, Altieri AH, McMillan WO, Leray M.
- 2021 ● **Rapid ecosystem-scale consequences of acute deoxygenation on a Caribbean reef**
[Nature Communications 12, 4522 \(2021\)](#) 
Johnson MD, **Scott JJ**, Leray M, Lucey N, Lucia Rodriguez L, Wied W, Altieri AH.
- 2020 ● **Intestinal microbes: an axis of functional diversity among large marine consumers**
[Proceedings of the Royal Society B: Biological Sciences 287:\(20192367\)](#) 
Scott JJ, Adam TC, Duran A, Burkepile DE, Rasher DB.
- 2020 ● **A Genus definition for Bacteria and Archaea based on a standard genome relatedness index**
[mBio 11\(2020\):e02475-19](#) 
Barco RA, Garrity GM, **Scott JJ**, Amend JP, Nealson KH, Emerson D.
- 2018 ● **Biological rejuvenation of iron oxides in bioturbated marine sediments.**
[The ISME Journal. 12\(2018\):1389-1394.](#) 
Beam JP, **Scott JJ**, McAllister SM, Chan CS, McManus J, Meysman FJ, Emerson D.

[Project website](#) & reproducible workflows from this paper. Note: I do my best to publish open access articles. Sometimes I fail. If you have issue accessing any of my papers, please email me and I will pass along a pdf.

[Project website](#) & reproducible workflows from this paper

[Project website](#) & reproducible workflows from this paper. Johnson, **Scott**, Leray, & Lucey contributed equally to the work.

[Project website](#) & reproducible workflows from this paper.

Editor's Pick

- 2017 ● Bringing microbial diversity into focus: high-resolution analysis of iron mats from the Lō'ihi Seamount.
[Environmental Microbiology. 19\(2017\):301-316.](#)
Scott JJ, Glazer BT, Emerson D.
- 2017 ● Physiological and ecological implications of an iron- or hydrogen-oxidizing member of the Zetaproteobacteria, *Ghiorsea bivora*, gen. nov., sp. nov.
[The ISME Journal. 11\(2017\):2624-2636.](#) 
Mori JF, **Scott JJ**, Hager KW, Moyer CL, Küsel K, Emerson D.
- 2017 ● Biogeography of mutualistic fungi cultivated by leafcutter ants.
[Molecular Ecology. 26\(2017\):6921-6937.](#)
Mueller UG, Ishak HD, Bruschi SM, Smith CC, Herman JJ, Solomon SE, Mikheyev AS, Rabeling C, **Scott JJ**, Cooper M, Rodrigues A.
- 2017 ● *In situ* estimates of iron-oxidation and accretion rates for iron-oxidizing bacterial mats at Lō'ihi Seamount.
[Deep Sea Research Part I: Oceanographic Research Papers. 126\(2017\):31-39.](#)
Emerson D, **Scott JJ**, Leavitt A, Fleming E, Moyer C.
- 2015 ● Microbial iron mats at the Mid-Atlantic Ridge and evidence that Zetaproteobacteria may be restricted to iron-oxidizing marine systems.
[PLoS One. 10\(2015\):e0119284.](#) 
Scott JJ, Breier JA, Luther III GW, Emerson D.
- 2015 ● Baleen whales host a unique gut microbiome with similarities to both carnivores and herbivores.
[Nature Communications. 6\(2015\):8285.](#) 
Sanders JG, Beichman AC, Roman J, **Scott JJ**, Emerson D, McCarthy JJ, Girguis PR.
- 2015 ● Microbial iron oxidation in the arctic tundra and its implications for biogeochemical cycling.
[Applied & Environmental Microbiology. 81\(2015\):8066-8075.](#) 
Emerson D, **Scott JJ**, Benes J, Bowden WB.
- 2015 ● Unique honey bee (*Apis mellifera*) hive component-based communities as detected by a hybrid of phospholipid fatty-acid and fatty-acid methyl ester analyses.
[PLoS One. 10\(2015\):e0121697.](#) 
Grubbs KJ, **Scott JJ**, Budsberg KJ, Read H, Balser TC, Currie CR.

- 2013 ● *Leucoagaricus gongylophorus* produces diverse enzymes for the degradation of recalcitrant plant polymers in leaf-cutter ant fungus gardens.
[Applied & Environmental Microbiology](#) 79(2013):3770-3778. 
Aylward FO, Burnum-Johnson KE, Tringe SG, Teiling C, Tremmel DM, Moeller JA, **Scott JJ**, Barry KW, Piehowski PD, Nicora CD, Malfatti SA.
- 2012 ● Metagenomic and metaproteomic insights into bacterial communities in leaf-cutter ant fungus gardens.
[The ISME Journal](#). 6(2012):1688-701. 
Aylward FO, Burnum KE, **Scott JJ**, Suen G, Tringe SG, Adams SM, Barry KW, Nicora CD, Piehowski PD, Purvine SO, Starrett GJ.
- 2011 ● The genome sequence of the leaf-cutter ant *Atta cephalotes* reveals insights into its obligate symbiotic lifestyle.
[PLoS Genetics](#). 7(2011):e1002007. 
Suen G, Teiling C, Li L, Holt C, Abouheif E, Bornberg-Bauer E, Bouard P, Caldera EJ, Cash E, Cavanaugh A, Denas O, Elhaik E, Fav MJ, Gadau J, Gibson JD, Graur D, Grubbs KJ, Hagen DE, Harkins TT, Helmkamp M, Hu H, Johnson BR, Kim J, Marsh SE, Moeller JA, Muoz-Torres MC, Murphy MC, Naughton MC, Nigam S, Overson R, Rajakumar R, Reese JT, **Scott JJ**, Smith CR, Tao S, Tsutsui ND, Viljakainen L, Wissler L, Yandell MD, Zimmer F, Taylor J, Slater SC, Clifton SW, Warren WC, Elsik CG, Smith CD, Weinstock GM, Gerardo NM, Currie CR.
- 2010 ● Microbial community structure of leaf-cutter ant fungus gardens and refuse dumps.
[PloS One](#) 5(2010):e9922. 
Scott JJ, Budsberg KJ, Suen G, Wixon DL, Balser TC, Currie CR.
- 2010 ● An insect herbivore microbiome with high plant biomass-degrading capacity.
[PLoS Genetics](#). 6(2010): e1001129. 
Suen G, **Scott JJ**, Aylward FO, Adams SM, Tringe SG, Pinto-Tomás AA, Foster CE, Pauly M, Weimer PJ, Barry KW, Goodwin LA.
- 2010 ● Monoculture of leafcutter ant gardens.
[PLOS One](#).5(2010):e12668. 
Mueller UG, **Scott JJ**, Ishak HD, Cooper M, Rodrigues A.
- 2009 ● Polymorphic microsatellite markers for the symbiotic fungi cultivated by leaf cutter ants (Attini, Formicidae).
[Molecular Ecology Resources](#). 9(2009):1391-1394.
Scott JJ, Kweskin MK, Cooper M, Mueller UG.
- 2008 ● Bacterial protection of beetle-fungus mutualism
Scott JJ, Oh DC, Yuceer MC, Klepzig KD, Clardy J, Currie CR.
[Science](#). 2008 322(5898):63.

It might seem ridiculous to include a paper where I am the 33rd author-and normally I would agree. However I collected the ants used to sequence the first leaf-cutter ant genome. I am quite proud of that. It also happens to be one of my favorite field stories.

See accompanying Perspective: *Bugs Bugs*. Berenbaum MR, Eisner T. 2008. [Science](#). 322:52-53.



RECENT TEACHING EXPERIENCE

2024
2022
2020
2019



STRI-McGill NEO Tropical Biology Field Course

Instructor & Data Specialist

📍 Panama

- Guide project design & implementation.
- Reproducible workflows using Quarto.
- Assist students with field work.
- Natural history of neotropical marine & terrestrial ecosystems.
- Snorkeling class for inexperienced students.

[Course tutorial website.](#)

Field sites incl. [Barro Colorado Island](#), [Bocas del Toro](#), [Ft Sherman Canopy Crane](#), [Pipeline Road Forests](#), [Agua Salud](#) & [Isla Coiba](#).

2018



Marine Microbiome Workshop

Creator, Organizer, & Host

📍 Bocas del Tora, Panama

From model organisms to ecosystems: scaling-up our understanding of host-microbe symbiosis in the sea.

- Conceived, created & designed workshop.
- Workshop logistics & organization.
- Led discussions & working groups.

I teach the way I learn. My goal is to create a venue where students can be curious, get their hands dirty, make mistakes, & explore. I'm here to help students see what's possible, not tell them what to do.

[Workshop website](#) & [publication written by participants](#).



MEDIA FEATURES

2024



[Entangled](#)

STRI-McGill field course weaves together natural history, social science, tropical ecology, and cultural exchange

2021



[Picture the Unseen](#)

A community of scientists put microbial ecosystems back together again using anvi'o

2021



[Caribbean asphyxiation](#)

Desperate brittlestars, suffocating corals and resilient microbes: First multidisciplinary data from an acute marine hypoxic event and its historical context

2018



[Eastern Tropical Pacific](#)

Marine expedition launched from new Coibita station

2016



[It's complicated](#)

Why study symbiotic relationships in Panama?

2013



Cover Photo for [Applied & Environmental Microbiology](#) 79(12)

2010



Cover Photo for [PLoS Genetics](#) Vol. 6(9)

I helped write this article. I performed all of the analyses and created all figures.



WEB PRODUCTS & WORKFLOWS

As part of my work I create data-driven workflows & presentations using R, Python, & an open-source scientific & technical publishing system called [Quarto](#). Below is a selection of web projects I have created for courses, workshops, publications, & presentations.

2024	● Istmobiome Project Reproducible bioinformatic workflows for the Istmobiome microbiome project.	Project Website & Reproducible Workflows
2022	● SWELTR Reproducible bioinformatic workflows for the study Soil Warming Experiment in Lowland Tropical Rainforest.	Reproducible Workflow
2022	● R Markdowng Fielduide Web project guide book for 2022 STRI-McGill NEO Tropical Biology Field Course.	Course
2022	● BocasBiome Reproducible bioinformatic workflows for the study <i>The gut microbiome stability of a butterflyfish is disrupted on severely degraded Caribbean reef habitats..</i>	Reproducible Workflow
2021	● Hypocolypse Reproducible bioinformatic workflows for the study <i>Rapid ecosystem-scale consequences of acute deoxygenation on a Caribbean reef.</i>	Reproducible Workflow
2020	● ProjectDIGEST Reproducible bioinformatic workflows for the study <i>Intestinal microbes: an axis of functional diversity among large marine consumers.</i>	Reproducible Workflow
2020	● Cacao Fermentation Talk about the microbiology of cacao fermentation.	Public Presentation
2020	● Rethinking the Diversity of Life Talk about understanding diversity through a molecular lens.	Public Presentation
2019	● How the Isthmus of Panama Changed the World Talk about how life changed on land & in the sea after the closure of the Isthmus of Panama.	Public Presentation
2018	● Marine Microbiome Workshop Web site for the first STRI/Moore Foundation Marine Microbiome Workshop.	Workshop

The source code for this cv is available [here](#). I made it with the R package [pagedown](#) and help from the Internet, especially this [repo](#).

Last updated on 2026-05-19.